

# ECO 334—DEMOGRAPHIC ECONOMICS OF DEVELOPING COUNTRIES

## Online Course

Winter 2018: January 2–20, 2018

Department of Economics  
Stony Brook University

### 1 Statement about Course Delivery/Modality

This is an asynchronous, online course, delivered in the BlackBoard Learning Management System (LMS). Students must be mindful of all course expectations, deliverables and due dates. All assignments and course interactions will utilize internet technologies. See the section on *Technical Requirements* for more information.

The official time zone for this course is the **Eastern Time Zone (EST)**<sup>1</sup>. All deadlines will be indicated using EST time, so please be mindful of that in case you are in another time zone.

### 2 Instructor & Contact Information

**Instructor:** Alejandro Melo Ponce

**E-mail:** alejandro.meloponce@stonybrook.edu

**Office:** N604, Social and Behavioural Sciences Building.

**Office Hours:** By appointment in person or via phone/video conference.

My preferred method of contact is via email. I will strive to respond emails as soon as possible but allow for 24 hours for a response. Please use your Stony Brook University email when contacting me as that is the preferred method of contact for the institution. When sending me an email, **please include in the subject line the keywords ECO334**, since that will insure that I will be able to locate your email and respond to it quickly.

### 3 Course Prerequisites

C or higher in ECO 320 or AMS 310 (Mathematical Statistics); C or higher in ECO 303 and 305 (Intermediate Microeconomic and Macroeconomic Theory).

### 4 Course Description

This course is concerned with the many links between population and economic development. By 2050, nearly all of the world's population growth—approximately an additional 2.5 billion people—will take place in the cities and towns of developing countries. Moreover, the coming decades will bring profound changes to the size and spatial distribution of the world's population. By 2050, 66 percent of the world's population

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<sup>1</sup>US East Coast time.

is expected to live in urban areas. Therefore, in order to understand the key links between demography and development, it is essential to study the many roles that urbanization plays in these complex phenomena.

This course will introduce you to some of the theoretical, empirical and computational tools that are used to study the *story* of urbanization and demographic change in developing countries. The main *characters* in this *story* are the 6.07 billion people living in such countries: who are they, what are they, where are they and how are they are the main questions we will try to answer. However, in describing this *story* is not enough to analyze the characters but also the *background* in which these characters interact. The reality nowadays for most of the people living in the developing world is one of combining the future—urban life—with the past—rural life—and the demographic changes that such transition entails. We will discuss and analyze issues related with this transition, namely growth, poverty, migration, fertility, life expectancy and mortality. Furthermore, we will discuss urban governance as one of the central challenges that the governments of developing countries face; national governments across the developing world are increasingly ceding responsibilities of basic service delivery to local and regional governments, which in many cases are ill-prepared to handle such responsibilities. Finally, if time permits, we will close the our analysis of this *story* by exploring the urban health risks posed by climate change, with an emphasis to the problems that urban-dwellers and their governments will have to confront in the near future.

#### 4.1 Course Objectives

At the end of this course, the student will be able to:

- Know some of the theoretical and empirical tools of modern demographics and economics to analyze relevant issues in the demographic transition of developing countries.
- Point out some of the main issues with urbanization as it pertains to developing countries and its impact on economic development and other social and demographic issues, like geographic mobility, fertility, life expectancy, etc.
- Introduce the students to spatial analysis in the R language and the tools of *Geographic Information Systems* (GIS).

#### 4.2 Main Topics

1. Introduction to the course: main issues in the demographic transition and urbanization of developing countries. Introduction to Spatial Analysis with the R language, introduction to GIS.
2. Urban demography, urbanization and economic development.
3. Fertility, age structure and the economy.
4. Urbanization and economic development.
5. Rural-urban linkages.
6. Urban poverty.
7. Urban health.
8. New actors in the Urban Scene.
9. Overview of Climate Change Adaptation.
10. Urban Displaced Populations.

## 5 Course Structure

The course is divided into five modules, approximately two per week for the duration of the Winter Session 2018.

Most of the modules are structured as follows:

1. A lecture, in the form of slides, which introduces the background of the topic that is discussed and the theoretical and empirical tools used to analyze that topic.
2. An exploratory exercise in which I provide the necessary code to implement some of the empirical tools discussed in the lecture in the R language.

Some modules may not include the R language component.

In each module, your main duties will be to take a look at the slides and at the R code if available. You will also need to participate on a discussion forum for each of the modules. Participating in the discussion forums is the most important part of the course.

During the course, you will have **three** empirical assignments which will ask you to carry out some analysis using R. These exercises will be heavily based on the explanatory exercises from the modules, all of them usually just require a small modification of the code that I provide to you or to carry out exactly the same analysis but with another country, for example. I will provide you with the data sets that you will need to carry out these assignments.

In each of the modules, I will also provide you with recommended readings if you want to acquire a deeper knowledge of the material. This material in principle is optional but you should try your best to do the readings for the topics that you find interesting. Doing this will be important since at the end of the course you will need to deliver an essay/research paper in which you discuss a particular topic of your interest. Please see below for more information.

## 6 Bibliography & Course Material

There are no required textbooks. The lecture notes, discussion forums and other content provided on Blackboard are in principle sufficient for the course. All of these material will be posted on BlackBoard <https://blackboard.stonybrook.edu/>.

However, for some topics I will recommend some additional reading from the following sources:

1. National Research Council. *Cities Transformed: Demographic Change and Its Implications in the Developing World*. Ed. by Mark R. Montgomery et al. Washington, DC: The National Academies Press, 2003. ISBN: 978-0-309-08862-6. DOI: 10.17226/10693. URL: <https://www.nap.edu/catalog/10693/cities-transformed-demographic-change-and-its-implications-in-the-developing>.
2. Michael P. Todaro and Stephen C. Smith. *Economic Development*. 12th. Pearson, 2015. ISBN: 9780133406788.
3. Debraj Ray. *Development economics*. Princeton University Press, 1998. ISBN: 9780691017068.

## 7 Course Assessment and Grading Scale

In this course, you will be assessed according to the following grading policy:

| ASSIGNMENTS                                    | % OF GRADING |
|------------------------------------------------|--------------|
| Attendance, Participation and Discussion       | 30%          |
| R empirical assignments (three of them)        | 30%          |
| Final essay/research paper (see details below) | 40%          |

## 7.1 Assignment Description

### 7.1.1 Attendance, Participation and Discussion (36%)

**Attendance** Attendance in an online course means **logging into the Blackboard on a regular basis and participating in the all of activities that are posted in the course.** This online course requires that you log in throughout the week for the duration of the course; each module will be available as we progress throughout the term and you will be responsible for logging and understanding the module's deliverables.

**Participation** Participation in this course is comprised of discussion forums that are required and graded. There are a total of **6 graded discussion forums**, one for each module, each one of them worth 5 points.

The first module is the only one in which you will have two forums, an Introduction forum and that module's topic forum. In the introduction forum you will need to introduce yourself to the rest of the class. The introduction forum is worth 5 points.

For each module, you will be required to post an original and thoughtful response to the topic of that module's forum. For some modules you will have to post an initial response before you start studying the material for that week's module and then post another comment after you have studied the material. In all discussion forums you must also post at least two responses to two different posts by your colleagues. I will be more specific in your tasks for each module's discussion forum in its respective BlackBoard description.

I will grade the weekly module forums according to the following rubric:

| Score | Knowledge                                | Understanding                                                                         | Writing Skills                                                               | Application/Analysis                                                                                                  | Peer responsiveness                                                                                             |
|-------|------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 5     | Comprehensive, in-depth and wide ranging | Outstanding ability to grasp concepts and Relate theory to practice                   | Excellent mechanics, sentence structure, and organization                    | Grasps inner relationship of concepts, Excellent use of a wide range of supporting material                           | Demonstrates an exceptional ability to analyze and synthesize student work, asks meaningful extending questions |
| 4     | Up to date and relevant.                 | High level of ability to conceptualize essential ideas and relate theory to practice. | Significant mechanics, structure and organization.                           | Demonstrates the ability to analyze and synthesize, independent analysis, good use of a range of supportive material. | Demonstrates good ability to analyze other student work, and ask meaningful extending questions.                |
| 3     | Relevant but not comprehensive.          | Some ability to conceptualize essential ideas and relate theory to practice.          | Some grammatical lapses, uses emotional response in lieu of relevant points. | Informed commentary with some evidence of genuine analysis; some supportive materials used.                           | Some ability to meaningfully comment on other student work and ask bridging questions.                          |
| 2     | Limited superficial knowledge/response.  | Limited ability to draw out concepts and relate theory to practice.                   | Poor grammar, weak communication.                                            | Some observations, some supportive evidence used.                                                                     | Lack of ability to comment on other students work and ask meaningful questions.                                 |
| 1     | Little relevance/some accuracy.          | Minimal awareness that external concepts exist.                                       | Lack of clarity, poor presentation of thinking.                              | Lacks evidence of critical analysis, poor use of supportive evidence.                                                 | Demonstrates poor ability to comment on other student work, asks no meaningful questions.                       |
| 0     | Did not participate in forum.            |                                                                                       |                                                                              |                                                                                                                       |                                                                                                                 |

Participation in the weekly forums is of extreme importance in this course, so no make-ups are available for the discussion forums.

**Discussion** You are also responsible to **comment a few of your colleagues' posts**. You are to conduct yourself professionally every time you express yourself-whether it's through the written or spoken word.

### 7.1.2 Assignments in R (30%)

You will have to deliver **three** assignments in R. These assignments are heavily based on the material discussed in the R exploratory exercise in each of the modules. I will mostly ask you, using the code that I provide in each of the modules, to modify it a little bit in order to explore a particular issue or to replicate the same exercise by using another database. Each of the assignments will correspond roughly to the material of two modules.

I will post these assignments on Blackboard and the deadline by which they need to be submitted.

I will open up an **ungraded discussion forum** pertaining to each of the assignments in which you will be able to discuss with your classmates and the instructor any questions you might have about the assignment. The purpose of this is to give incentives for a collaborative learning environment. You are not required to post anything in this type of discussion forum but I strongly recommend that you participate, since learning is not only done through ourselves but also through our classmates and peers.

### 7.1.3 Final essay/research paper (40%)

Your final task for the course will involve writing an essay about a topic from your own choice that deals with a demographic/economic problem faced by a developing country. The topic of the essay should not be broadly defined but very specific. In order to do this, you will need to conduct a reference and literature review in order to delineate very precisely what you will write about. In this part, it will come in handy the optional references that I will post about each of the topics.

When writing your essay you will need to:

1. Give a precise description of the topic you are dealing with.
2. Put the topic in context with respect to the research/literature previously done on such topic.
3. Find some empirical data that will help you in carrying out the analysis of your chosen topic.
4. Propose the theoretical and analytical background with which you will analyze your topic.
5. Describe the analysis that you carried out. I expect at the minimum, a description of the data set that you are using and well constructed argument about why this data set illuminates the analysis of the topic that you chose. In constructing this argument, you must connect the theoretical and empirical tools that you identified in the previous point with the data that you are using. The best papers will of course try to carry out the empirical analysis using the tools learned in R.

The essays will have to conform with the appropriate formatting guidelines for academic writing, namely, the discourse should be formal and precise and you will need to properly cite and give credit for the ideas of other authors whose information you will provide in a references list. The length of the essays will be required to stay between 3000–8000 words, not including figures, title pages or the reference list. I will provide more details on the week before the essay is due.

#### 7.1.4 Late/Missed Assignments Policy

For the discussion forum assignments, the deadlines are strict, no late posts will be accepted under any circumstance. For the R assignments, no late assignments will be accepted beyond two days after the original due date. Late assignments will receive a penalty of 2 points less for each day they are late. For the final essay, no late submissions will be admitted.

**Important:** If at any point you should encounter any technical issues, please send an email to the help desk at [spd\\_online@stonybrook.edu](mailto:spd_online@stonybrook.edu); You should also copy me on this email so that I am aware of the situation and if possible include a screen-shot of the issue.

#### 7.1.5 Grades and Grading Scale

Assignment of letter grades is based on the total amount you obtained in all course assignments. Your numerical grade will be converted to a letter grade based on the following table.

| Letter Grade | Score Achieved |
|--------------|----------------|
| A            | [95, 100]      |
| A-           | [90, 95)       |
| B+           | [87, 90)       |
| B            | [83, 87)       |
| B-           | [77, 83)       |
| C+           | [70, 77)       |
| C            | [63, 70)       |
| C-           | [57, 63)       |
| D+           | [50, 57)       |
| D            | [40, 50)       |
| F            | [0, 40)        |

## 8 Academic Policies

### 8.1 Academic Integrity Statement

Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Faculty are required to report any suspected instances of academic dishonesty to the Academic Judiciary. Faculty in the Health Sciences Center (School of Health Technology & Management, Nursing, Social Welfare, and Dental Medicine) and School of Medicine are required to follow their school-specific procedures. For more comprehensive information on academic integrity, including categories of academic dishonesty, please refer to the academic judiciary website at <http://www.stonybrook.edu/uaa/academicjudiciary/>

### 8.2 Critical Incident Management

Stony Brook University expects students to respect the rights, privileges, and property of other people. Faculty are required to report to the Office of Judicial Affairs any disruptive behavior that interrupts their ability to teach, compromises the safety of the learning environment, or inhibits students' ability to learn. Faculty in the HSC Schools and the School of Medicine are required to follow their school-specific procedures.

### **8.3 University Student Conduct Code**

It can be found at: <http://studentaffairs.stonybrook.edu/ucs/docs/universitystudentconductcode.pdf>

### **8.4 ADA & Disability Support Services (DSS) Statement**

The Rehabilitation Act of 1973 - Section 504 applies to all postsecondary educational programs that receive federal assistance. Reasonable accommodations and academic assistance are provided to students with disabilities registered with the Disability Support Services, ECC (Educational Communications Center) Building, room 128, (631) 632-6748. They will determine with you what accommodations, if any, are necessary and appropriate. All information and documentation is confidential. For procedures and information go to the following website: <http://www.stonybrook.edu/ehs/fire/disabilities>

### **8.5 Course Materials and Copyright Statement**

Course material accessed from Blackboard, SB Connect, SB Capture or a Stony Brook Course website is for the exclusive use of students who are currently enrolled in the course. Content from these systems cannot be reused or distributed without written permission of the instructor and/or the copyright holder. Duplication of materials protected by copyright, without permission of the copyright holder is a violation of the Federal copyright law, as well as a violation of Stony Brook's Academic Integrity and Student Conduct Codes.

## **9 Communication Competency**

You convey a certain image of yourself every time you express yourself-whether it's through the written or spoken word. As such, successful professionals (students, managers, employees) must have effective written and oral communication skills. Therefore, communication errors (such as improper grammar, improper sentence and paragraph structure, misspelling and incorrect punctuation) are unacceptable in coursework throughout. Faculty members also consider communication competency when evaluating student performance in addition to coursework.

### **9.1 Email and Internet**

You must have an active Stony Brook University e-mail account and access to the Internet. All instructor correspondence will be sent to your SBU e-mail account. Please plan on checking your SBU email account regularly for course related messages. To log in to Stony Brook Google Mail, go to <http://www.stonybrook.edu/mycloud> and sign in with your NetID and password.

## **10 Net-Etiquette Guidelines**

The following guidelines for participation in the Threaded Discussion Area and Chat rooms.

- Remember that the absence of face-to-face communication it's easy to misunderstand what is being said;
- Carefully review and read materials that you receive electronically to ensure that you fully understand the message;

- Be sure to carefully re-read and understand what you will be sending in order to ensure that you are not misunderstood by anyone;
- Avoid cluttering your messages with excessive emphasis (stars, arrows, exclamations);
- If you are responding to a message, either include the relevant part of the original message in your message, or make sure refer to the original's contents so as to avoid confusion;
- Be specific and clear, especially when asking questions;
- If your messages can be typed in UPPER and lower case, please use the two appropriately instead of all UPPERCASE characters. This gives the appearance of shouting and makes the message less readable;
- Remember that not all readers have English as their native language, so make allowance for possible misunderstandings and unintended discourtesies;
- Do not abuse new users of computer networks for their lack of knowledge;
- Follow the same standards of politeness as you do in any other aspect of your life.

## 11 Technical Requirements

This course requires that you have access to the Internet. You are responsible for having a reliable computer and internet connection throughout the course. You will need to have access to, and be able to use the following software:

- A web browser (for instance, Google Chrome, Mozilla Firefox or Internet Explorer)
- Adobe Acrobat Reader (free)
- Adobe Flash Player (free)
- Microsoft Word and Microsoft Excel (free to Stony Brook University students, go to <https://it.stonybrook.edu/software/title/microsoft-office>)
- Rstudio and an installation of the R language. Obtaining this software is free and we will go over how to install it during the course.

*Please note!* You **will** be limited if you expect to complete your work on a smart phone or tablet. It will not be possible to submit the Microsoft Word files required for your quiz/exam assignments, neither to view some of the embedded material in Blackboard. This course uses Blackboard for the facilitation of communications between faculty and students, submission of assignments, and posting of grades. The Blackboard Course Site can be accessed at <https://blackboard.stonybrook.edu>.

## 12 Getting Technical Help

### 12.1 Campus Network or Blackboard Outage

When access to Blackboard is not available for an extended period of time (greater than one entire evening—6PM till 11PM EST) you can reasonably expect that the due date for assignments will be changed to the next day (assignment still due by midnight).

## 12.2 Help-Desk

Go to the menu item “SPD Online Support” and go to the Discussion Board within the support site. There you will find a Technical Questions and Answers Forum. Post your question there, and someone from the SPD Online office will respond. Alternatively, you could call the SPD Online office at 631.632.9484 between the hours of 9:00am and 5:00pm, Monday through Friday or e-mail [spd\\_online@stonybrook.edu](mailto:spd_online@stonybrook.edu). For assistance after 5:00 PM EST or over the weekend, please contact the Open SUNY Technical Support, information can be found at <http://open.suny.edu/support/contact-us/current-students/>

## 12.3 Getting Help with Bb Learning Management System (LMS)

First contact SPD Online at [spd\\_online@stonybrook.edu](mailto:spd_online@stonybrook.edu) or 631-632-9484. After hours contact Open SUNY Technical Support, see above for contact information. Additionally students that need help with Bb can contact the TLT Student Help Desk by calling (631) 632-9602, emailing [helpme@stonybrook.edu](mailto:helpme@stonybrook.edu); more information is available via Stony Brook IT: <http://it.stonybrook.edu/services/blackboard/section-6706>. Frequently asked questions about the Bb LMS along with tutorials are available here: <http://it.stonybrook.edu/services/blackboard/navigate-manage>.

# 13 Academic Support

## 13.1 SPD Subject Guides

The SPD Subject Guide is a library website specifically designed to assist SPD students with their research. It contains all of the information referenced on this Blackboard page as well as recommendations for specific databases and a live librarian chat feature. Take a look: <http://guides.library.stonybrook.edu/spd>

## 13.2 Student Support for Online Learning

<http://www.stonybrook.edu/commcms/onlineed/student.html>

## 13.3 Library Instruction Website

<http://library.stonybrook.edu/workshops-this-week-citation-skills-worldcat-and-end-note-the-hsc/>

## 13.4 SBU Library Research Guides and Tutorials

<http://library.stonybrook.edu/research/research-basics/>

# 14 Course Schedule

This weekly schedule is designed in a way that supports students with the time allocation and also gives them a benchmark of progress. Feel free to work at your own pace, but make sure you meet all deadlines timely. As explained in the Course Description section above, this course is divided into **Six Modules**, one for each week, as follows:

- **Module 1** (January 2–4)
  - Introduction: main issues, Spatial Analysis, GIS and R.

- **Module 2** (January 5–7)
  - Urbanization & Fertility, Mortality and Life Expectancy
- **Module 3** (January 8–10)
  - Economic Links Between urbanization and Development & Rural Urban Linkages.
- **Module 4** (January 11–13)
  - Urban Health
- **Module 5** (January 14–16)
  - Introduction to Climate change
- **Final essay**
  - Start thinking about the topic that interests you since the beginning of the class.
  - You will have the week from January 14–20 to focus almost exclusively on researching and writing your essay, since the last module is comparatively easier and shorter to the other modules.
  - The deadline to deliver the essay is **January 20, 11:59PM EST**. No late submissions will be admitted.

## 15 Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on BlackBoard.